Zhengyang Yu

Software Engineering Student

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA
Expected Graduation: June 2028; GPA: 4.0/4.0
Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*
- Designed and built a modular C++/FreeRTOS application that authenticates multiple accounts, retrieves structured data over HTTPS, and presents live state through an interactive physical display.
- Implemented concurrent network workers, adaptive polling, persistent configuration, and bounded structured logs; investigated failures spanning authentication, state management, networking, and rendering.
- Separated service and UI responsibilities, documented the custom server API, and verified changes through repeatable builds, device deployment, and hardware testing.

2025 **HTML to ePub Converter**, *Personal Project*
- Engineered a Python data-processing pipeline that parses HTML and XML, validates document structure, embeds metadata, and produces standards-compliant ePub files.
- Improved maintainability through modular transformations, structured error handling, and repeatable validation of generated output.

2024 **Home Assistant Presence Plugin**, *Personal Project*
- Built a Python integration that models room occupancy from entry and exit sensor events and exposes real-time state for home automation.
- Designed an event-driven REST API workflow with explicit state transitions to support testing, troubleshooting, and reliable smart-lighting behavior.

Skills

Programming	C++, Python, Java, C
Backend & Data	HTTP/HTTPS, REST APIs, JSON, XML, data processing, concurrent and event-driven systems
Engineering	Object-oriented design, data structures, algorithms, software testing, debugging, API design
Delivery	Requirements analysis, technical documentation, structured logging, configuration management
AI Development	Codex-assisted codebase exploration, implementation, refactoring, debugging, and verification
Tools	Git, Docker, Linux, PlatformIO, VS Code

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch
Supported community members with computer troubleshooting and taught basic software skills.